Do Now · retrieval
Records, lists and tuples

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does names[2] access in a 1D array?

Q2

In a C# 2D array organised as rows and columns, which dimension does GetLength(1) measure?

Q3

Which structure from last lesson has a fixed size once created?

Q4

Why should a student's name, candidate number and score be stored as separate related fields rather than one primitive value?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Records group fields — CODE:
public record Student(string Name, int CandidateNumber, int Score);
Student pupil = new Student("Amelia", 1037, 72);
Console.WriteLine(pupil.Name); — A record represents one entity using several related fields. In C#, this positional record creates a named Student type with properties. Those properties use init, so use a with expression to create an updated copy.

Core idea 2

Lists can change size — CODE:
List<string> names = new List<string>();
names.Add("Amelia");
names.Add("Ben");
names.Remove("Ben"); — A list is ordered like an array, but its size can change while the program runs.

Core idea 3

Tuples group a few values — CODE:
var result = (Name: "Amelia", Score: 72);
Console.WriteLine(result.Name);
Console.WriteLine(result.Score); — A tuple groups values without declaring a reusable Student type. It is especially helpful when a method returns several results.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Choose by behaviour — CODE:
fixed size? → array
one item / fields? → record
grow or shrink? → list
small grouped set? → tuple — The exam may ask why one structure suits a context better than another.

Vocabulary · use the correct technical terms

record — A collection of related fields describing one item.

field — One named piece of data inside a record.

list — A variable-size ordered collection.

tuple — A small fixed grouping of values.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Build and use a C# record

Follow the worked method

1. Declare the record type — CODE:
public record Student(string Name, int CandidateNumber, int Score);

2. Access a property — CODE:
Console.WriteLine(pupil.Score); // 72

3. Update using with — CODE:
Student updated = pupil with { Score = 80 };
Console.WriteLine(updated.Score); // 80

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Fill the same code
CODE:
names.?("Daisy");
names.?("Ben");

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Create and edit a list

Create a List<string> of four names. Add a fifth, remove one name, then output every remaining item.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Choose and code a record

Write short comments explaining when you would choose an array, list, record-like structure and tuple.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A C# List<T> can grow and shrink while the program is running.

Use key vocabulary from memory

1. Explain record in your own words.

2. Explain field in your own words.

3. Explain list in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.